

Evaluation of the ONH Expert Connector Search

Overview

As part of a benchmark user test for the Oslo Nye Høyskole (ONH) Expert Connector, I evaluated the retrieval-augmented generation (RAG) search engine that links user queries to staff expertise. The goal was to determine whether the application correctly identifies the most relevant experts, ranks them appropriately and provides attributable evidence. To conduct the evaluation I first built a **gold-standard staff-expertise reference**, drawing on the official **Finn en ekspert** page and staff profile pages on the ONH website as well as available publications on the NVA database. This reference lists dozens of topics (e.g. *Arbeidsmiljø*, *Anatomi*, *Digitalisering*, *Afrikas Horn*) and maps them to staff members who have declared expertise in those areas ¹ ². I then crafted a diverse set of 64 search queries covering psychology/HR, economics and administration, health/medicine, international relations and cross-cutting themes. Each query was executed in the ONH Expert Connector, and I recorded the actual results alongside the expected experts, missing experts, unexpected results and evidence quality in a structured log (see `onh_test_log.md`).

Staff-Expertise Reference (Gold Standard)

The **Finn en ekspert** page groups ONH expertise into categories such as **Psykologi og HR**, **Økonomi og administrasjon**, **Helse, medisin og ernæring** and **Statsvitenskap og internasjonale relasjoner**. Each topic on that page points to one or more staff profiles, providing a reliable mapping between expertise and individuals. For example, under psychology/HR, topics like *Arbeidsmiljø* and *Arbeidspsykologi* map to Anniken Grønstad, Ingvild Sagberg, Linda Mari Seem Sundvik and Mona Berthelsen ¹; *Kritisk psykologi* maps to Erik Nakkerud, Sandra Gabrielsen and Joshua Phelps ³; and *Motivasjon* and *AI i utdanning og vurderingsformer* map to Eirik Kjelstrup. In the economics category, *Digitalisering* is assigned to Kateryna Reiby and *Endringsledelse* to Johan Bergh ⁴. In health/medicine, topics such as *Anatomi* and *Cellebiologi* list Stian Walle Larsen, Suvi Sagbraaten, Elin Lund Transeth and Ulrike Liisberg ⁵, while *Diabetes* and *Epidemiologi* map to Dagfinn Aune. For international relations, *Afrikas Horn* and *Autoritære regimer* map to Kjetil Tronvoll, *Spionasje* to Stig Stenslie and *Amerikansk politikk* to Hilde Eliassen Restad ².

This reference enabled me to identify which staff should appear for each query and provided justification for expected rankings. Where additional context was needed, I consulted staff profile pages to confirm areas of expertise (e.g. Eirik Kjelstrup's profile highlights his work on decision-making, emotions and stress ⁶).

Summary of Search Performance

Strengths

- **Correct retrieval for some well-defined topics.** When queries exactly matched the topic names on the official list, the system often returned the correct expert(s) with appropriate ranking. Examples include:
 - *arbeidspsykologi* → Mona Berthelsen, Linda Mari Seem Sundvik and Brita Bjørkelo appear, matching the official list with correct ordering (severity *Low*).
 - *digitalisering* and *anatomy/cell biology/biochemistry* queries returned Kateryna Reiby, Suvi Sagbraaten/Stian Walle Larsen or Øyvind Jacobsen with high precision 4 5 .
 - Several marketing queries (*digital markedsføring, finans, regnskap, merkevareledelse, kampanjeplanlegging*) produced the expected lecturers in top positions.
- In international relations, topics like *Afrikas Horn* and *Spionasje* correctly surfaced Kjetil Tronvoll and Stig Stenslie respectively, demonstrating the model can retrieve some niche geopolitical expertise.
- **Support for Norwegian queries and compound nouns.** Many queries used Norwegian compound phrases (e.g. *beslutningspsykologi, livsstilssykdommer*). The system generally tokenised these correctly, matching them to staff keywords and returning relevant experts (e.g. Eirik Kjelstrup for *beslutningspsykologi*, Ella Marie Heyerdahl for *livsstilssykdommer*).

Key Failure Modes

Despite some successes, numerous systematic issues emerged:

- **High false-negative rate for many topics.** In 25 out of 64 queries the primary expected expert was missing entirely, including *alkoholmisbruk ungdom* (missing Anne Kjeldsen), *kritisk psykologi* (missing Erik Nakkerud and co.), *beslutningsbias*, *stressmestring*, *gravitetstrening*, *radikalisering*, *FN* and *autoritaere regimer*. These failures suggest that the retrieval stage is not indexing or matching the official keywords effectively.
- **Frequent false positives and category leakage.** Many queries returned staff from unrelated fields. For instance, *arbeidsmiljø og hybridarbeid* produced a long list of names from international relations and nutrition; *spionasje* and *fred og konflikt* included political scientists not connected to espionage; and *diabetes* returned a psychologist (Line Wisting) before the epidemiologist Dagfinn Aune. The presence of Beate Mangset (a health ethicist) in numerous unrelated queries (e.g. *kommunikasjonstrening, arbeidskonflikter, styrketrening*) indicates the ranking model is overly influenced by generic terms like “veiledning” found in many profiles.
- **Poor ranking within retrieved candidates.** Even when the expected expert was retrieved, they were often not ranked first. E.g. *AI i utdanning* placed Eirik Kjelstrup fourth behind two digital-marketing lecturers; *endringsledelse* placed the actual course leader Johan Bergh below Sandra Gabrielsen; *komparativ politikk* put Henry Allen ahead of the comparativist Jonas Bergan Dræge. The algorithm seems to weight generic similarity or recency over domain relevance.

- **Evidence/citation mismatch.** The RAG system surfaces evidence snippets next to each expert, but these are sometimes generic (e.g. listing courses taught) rather than directly supporting the query. For example, the evidence for Eirik Kjelstrup in *AI i utdanning* referenced his psychological research rather than his AI work; some results displayed tags not obviously related to the query. In several cases (e.g. *FN, bistand*) the sole result had no connection to the query at all.
- **Limited handling of synonyms and language variants.** Queries using synonyms, abbreviations or adjacent phrases often failed. *Kommunikasjonstrening* (communication training) and *teambygging* (team building) did not retrieve the organisational psychology lecturers, while *autoritaere regimer* returned a comparativist instead of the authoritarian-regime specialists Kjetil Tronvoll and Stig Stenslie. Abbreviations such as *FN* (United Nations) matched “Folkehelse ernæring” rather than the intended international-relations term.

Patterns of Errors

1. **Keyword leakage across categories:** The search tends to match keywords at surface level, ignoring disciplinary context. Staff who mention “stress” or “veiledning” in any context were retrieved for queries about stress management or team building, irrespective of their field. Similarly, nutrition lecturers appeared for UN-related queries because their tags contained two letters matching the acronym.
2. **Insufficient use of official topic mapping:** The RAG pipeline does not appear to prioritise the authoritative mapping between topics and experts provided on the ONH website. Incorporating this structured knowledge as a high-weight signal could dramatically improve recall and ranking.
3. **Lack of query expansion:** Many queries could benefit from synonym expansion (e.g. *beslutningsbias* → *decision bias*) and morphological normalization to capture variations (e.g. *arbeidsmiljø* vs *arbeidsmiljø og hybridarbeid*). The current system seems brittle to such variation.
4. **Minimal filtering of evidence snippets:** The sources shown often consist of full profile descriptions. Shorter, targeted excerpts that mention the query terms would improve user trust and highlight relevance. In some cases, the evidence for irrelevant experts did not contain the query at all.

Recommendations

1. **Incorporate the official “Finn en ekspert” mapping into the retrieval pipeline.** Use the structured list of topics and associated staff as a knowledge base. When a query matches or semantically relates to a listed topic, those experts should be given a very high prior probability. This would prevent obvious misses such as *alkoholmisbruk ungdom* and *kritisk psykologi*.
2. **Improve semantic search with synonym and abbreviation handling.** Implement query expansion to include synonyms, morphological variants and common abbreviations. For example, map *FN* to “United Nations”, *autoritaere regimer* to “authoritarian regimes”, and recognise that *kommunikasjonstrening* relates to organisational communication training. This could be achieved by enriching the embedding model with a Norwegian thesaurus or manually curated synonym lists.
3. **Apply category-aware filtering and weighting.** Each staff member belongs to one or more categories (psychology/HR, economics, health, international relations). When a query clearly falls within a category, experts from unrelated categories should be heavily down-weighted. This would reduce false positives such as nutrition lecturers appearing in political science queries.

4. **Re-train ranking with relevance feedback.** Use the gold-standard dataset constructed here (queries with expected experts) to train or fine-tune the ranking model. The objective should prioritise recall of true experts and penalise the elevation of irrelevant candidates. Human-in-the-loop evaluations could iteratively refine weights.
5. **Improve evidence snippet extraction.** The RAG system should return concise passages from staff profiles or publications that explicitly mention the query topic. This may involve chunking profiles into smaller segments and scoring them against the query. Irrelevant evidence should be filtered out.
6. **Support multi-language queries.** Although Norwegian queries were accepted, some English or mixed-language queries (e.g. "cognitive remediation therapy") performed poorly. Adding multilingual embedding models and translation layers could expand coverage to English terminology used by students and faculty.

Suggested Benchmark Structure for Regression Testing

- **Dataset:** Use the 64 queries and their expected experts provided in `onh_test_log.md` as the baseline benchmark. For each query, include the list of expected experts and a severity rating reflecting the impact of missing them.
- **Metrics:** Measure recall (percentage of expected experts retrieved), precision (percentage of retrieved experts that are relevant) and ranking quality (e.g. mean reciprocal rank of the top expected expert). Also assess evidence quality by checking that returned snippets mention the query terms.
- **Regression procedure:** After each update to the search algorithm, re-run the benchmark queries and record metrics. Highlight regressions where previously correct queries now fail or ranking deteriorates. Use the severity ratings to prioritise fixes.

Conclusion

The ONH Expert Connector shows promise but currently struggles to consistently retrieve and rank the most relevant experts. Reliance on loose keyword matching leads to high false-negative and false-positive rates, and the ranking often misplaces key experts beneath unrelated staff. By integrating the authoritative expertise mapping from the ONH website, enhancing semantic understanding and implementing category-aware filtering and better evidence extraction, the system can significantly improve both precision and user trust. The comprehensive benchmark developed here provides a foundation for iterative improvements and future regression testing.

1 2 3 4 5 Finn en ekspert | Ekspertene ved ONH | Oslo Nye Høyskole
<https://oslonyehoysskole.no/Forskning/Finn-en-ekspert>

6 Eirik Kjelstrup | Oslo Nye Høyskole
<https://oslonyehoysskole.no/om-oss/Ansatte/Eirik-Kjelstrup>